Unit 2 FRQ Set 2: Coaching Guide

Use this in two passes. **Pass 1:** for any part you couldn't finish, read only the 'Coach' questions and try again (a different colour pen). **Pass 2:** grade both attempts against the rubric.

Question	Max	Attempt 1 (timed)	After coaching
Q1 MR	10		
Q2 TBR	12		
Q3 LAB	10		
Q4 QQT	8		
Total	**40**		

Rubrics follow the College Board's style ('1 point for…'). They're written for practice. They aren't official scoring guidelines, but they test the same skills.

Question 1: Mathematical Routines

10 points

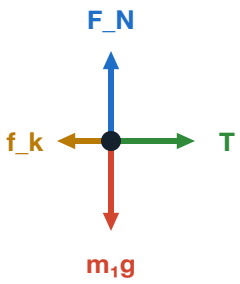
Block 1 (mass m_1 = 4.0 kg) slides on a rough horizontal table (μ_k = 0.25). It is tied by a light string over an ideal pulley to Block 2 (mass m_2 = 2.0 kg), which hangs. Block 1 moves toward the pulley.

Part A (2 pts)

COACH: ASK YOURSELF

- Count the contacts on Block 1: the table (gives TWO forces: which two?) and the string. Plus gravity.

A FULL-CREDIT ANSWER



Four forces: T toward the pulley, f_k opposite the motion, F_N up, m_1g down (F_N and m_1g equal in length).

All four forces present, correct directions	1 pt
Labelled, no extra forces; $F_N = m_1g$ in length	1 pt

WHERE POINTS LEAK Drawing m_2g on Block 1. Block 2's weight acts on Block 2, and it reaches Block 1 only through the tension.

Part B (3 pts)

COACH: ASK YOURSELF

- System along the string: which force drives the system, and which one resists it?
- Is the tension internal or external to the two-block system?

A FULL-CREDIT ANSWER

System (both blocks, along the string): $\Sigma F = (m_1 + m_2)a$.

Driving: m_2g. Resisting: $f_k = \mu_k m_1 g$. T is internal.

$a = (m_2 - \mu_k m_1)g / (m_1 + m_2)$

Newton's 2nd law for the system, or for each block with T eliminated	1 pt
Correct friction $\mu_k m_1 g$	1 pt
Correct expression	1 pt

WHERE POINTS LEAK Dividing by m_2 only. Every mass that moves gets accelerated, so all of them go in the denominator.

Part C (2 pts)

COACH: ASK YOURSELF

- For T, isolate the simpler block. Block 2 has only two forces.

A FULL-CREDIT ANSWER

(i) $a = (2.0 - 0.25 \cdot 4.0)(9.8)/6.0 = (1.0)(9.8)/6.0 =$ **1.6 m/s²**.

(ii) Block 2 (down +): $m_2g - T = m_2a \rightarrow T = 2.0(9.8 - 1.63) =$ **16 N**. It must be less than $m_2g = 19.6$ N ✓

Correct a with units	1 pt
Correct T with units	1 pt

WHERE POINTS LEAK Thinking $T = m_2g = 19.6$ N. That would mean Block 2 isn't accelerating.

Part D (3 pts)

COACH: ASK YOURSELF

- Put $m_1 = 8.0$ kg into your part B expression. What do you get?
- Zero acceleration while moving means what kind of motion? (Newton's 1st law.)

A FULL-CREDIT ANSWER

From part B: $a = (2.0 - 0.25 \cdot 8.0)g / 10 = 0$. The driving force $m_2g = 19.6$ N exactly balances kinetic friction $\mu_k m_1 g = 19.6$ N. The net force is zero, so by Newton's 1st law the blocks **keep moving at constant speed** (the speed they had after the push).

Constant velocity (keeps moving, doesn't speed up or stop)	1 pt
Uses the part B result, or shows the forces balance, with numbers	1 pt

Links ΣF = 0 to constant velocity (1st law) **1 pt**

WHERE POINTS LEAK Saying 'it stops'. ΣF = 0 doesn't mean at rest. (If the blocks started from rest, static friction could hold them, but they were already moving.)

THINK FURTHER What's the largest m_1 for which the blocks still speed up? ($m_2 > \mu_k m_1$, so $m_1 < 8.0$ kg.)

Question 2: Translation Between Representations

12 points

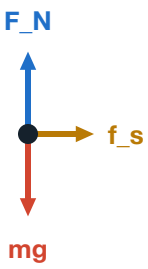
A crate of mass m sits on the flat bed of a truck. The coefficient of static friction between the crate and the bed is μ_s, and the coefficient of kinetic friction is μ_k ($\mu_k < \mu_s$). The truck starts from rest and speeds up with a constant acceleration a toward the right. The crate does not slide.

Part A (3 pts)

COACH: ASK YOURSELF

- The crate speeds up to the right along with the truck. Which force on the crate points right? It has to come from something the crate touches.
- Which way does the crate 'want' to slide relative to the bed? Friction opposes THAT.

A FULL-CREDIT ANSWER



Static friction points **forward** (right). It's the only horizontal force, so it's what accelerates the crate. Without friction, the crate would stay behind while the bed moved forward under it.

Gravity down and normal up, equal lengths	1 pt
Static friction forward, in the direction of the acceleration	1 pt
No extra forces (no 'force of the truck', no backward 'inertia force')	1 pt

WHERE POINTS LEAK Drawing friction backward. Students think 'friction opposes motion'. It opposes SLIDING between the surfaces, and here it drags the crate forward.

Part B (3 pts)

COACH: ASK YOURSELF

- At a_{max}, static friction is at its limit: f = ?
- Does m survive to the final answer?

A FULL-CREDIT ANSWER

Horizontal: $f_s = ma$. Vertical: $F_N = mg$.

At the limit: $\mu_s mg = ma_{max} \rightarrow \mathbf{a_{max} = \mu_s g}$

Newton's 2nd law with friction as the net force	1 pt
Uses $f_{s,max} = \mu_s F_N = \mu_s mg$	1 pt

Correct expression (m cancels)	1 pt

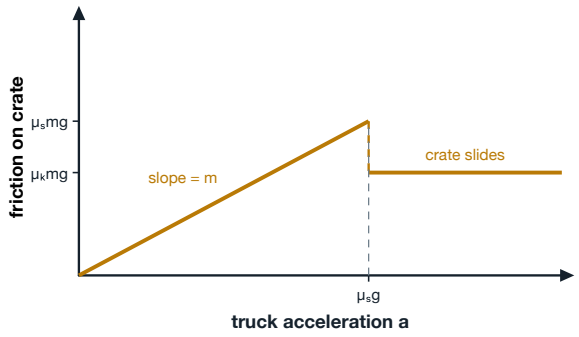
WHERE POINTS LEAK Writing $a_{max} = \mu_s mg/m$ and not simplifying. It's still right, but you'll miss the 'mass cancels' insight you need for part D.

Part C (4 pts)

COACH: ASK YOURSELF

- Below a_{max}: f = ma. What shape is that?
- Above a_{max}, the crate slides. Which kind of friction is it now, and does it depend on a?

A FULL-CREDIT ANSWER



Straight line through the origin for $a < a_{max}$	1 pt
Peak value $\mu_s mg$ at $a = \mu_s g$ (labelled)	1 pt
Drops to a lower constant value ($\mu_k mg$) beyond a_{max}	1 pt
Horizontal and constant after the drop	1 pt

WHERE POINTS LEAK Continuing the line upward past a_{max}. Once it slides, friction is $\mu_k mg$ no matter how hard the truck accelerates.

Part D (2 pts)

COACH: ASK YOURSELF

- Look at your part B expression. Is m in it?

A FULL-CREDIT ANSWER

Stays the same. The heavier crate gets twice the maximum friction ($\mu_s(2m)g$) but also needs twice the force for the same acceleration. From part B, $a_{max} = \mu_s g$ doesn't depend on mass.

Stays the same	1 pt
Justification: maximum friction and required force both scale with m, consistent with $a_{max} = \mu_s g$	1 pt

WHERE POINTS LEAK Saying 'heavier is harder to move, so it slides first'. That ignores the fact that the friction grows too.

THINK FURTHER The truck brakes instead. Which way does friction on the crate point now? (Backward. It still points in the direction of the crate's acceleration.)

Question 3: Experimental Design and Analysis 10 points

Students want to determine the inertial mass of a cart *without* using a balance. They have a low-friction track, the cart, a force sensor that can pull the cart horizontally, and a motion sensor that records the cart's velocity as a function of time.

Part A (2 pts)

COACH: ASK YOURSELF

- Newton's 2nd law links which three quantities? Which one will you control, and which will you measure?
- The motion sensor gives v(t). How do you get a from that?

A FULL-CREDIT ANSWER

1. Pull the cart along the level track with the force sensor, keeping the reading as steady as possible. Record F.
2. Use the motion sensor's v–t graph and take its slope as the acceleration a.
3. Repeat for at least 5 different pulling forces, with 3 trials each, and average.

Varies the applied force and measures it with the force sensor	1 pt
Finds a from the slope of the v–t data; repeated trials or several forces	1 pt

WHERE POINTS LEAK 'Measure the acceleration' with no HOW. Graders need the method: the slope of v vs. t.

Part B (2 pts)

COACH: ASK YOURSELF

- F = ma. If you put F on the vertical axis and a on the horizontal, what is the slope?

A FULL-CREDIT ANSWER

Graph **F (vertical) vs. a (horizontal)**. Because F = ma, the slope is the mass m. (Or graph a vs. F, where the slope is 1/m.)

A linearizing choice of axes (F vs. a or a vs. F)	1 pt
Correctly states the slope = m (or 1/m)	1 pt

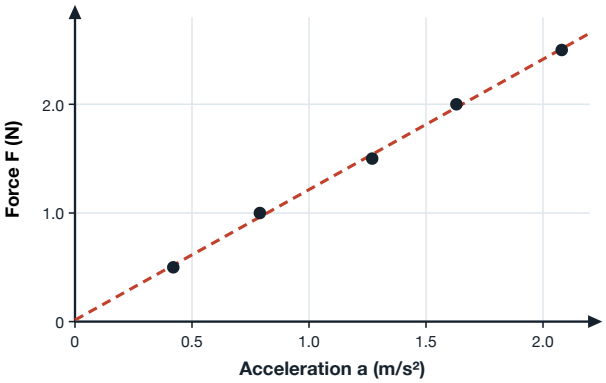
WHERE POINTS LEAK Getting the relationship between the slope and the mass backward.

Part C (4 pts)

COACH: ASK YOURSELF

- Same 4-point checklist as always: labels + units, scale, points, best-fit line.

A FULL-CREDIT ANSWER



Axes labelled with units	1 pt
Appropriate linear scale	1 pt
Points plotted correctly	1 pt
Best-fit line	1 pt

WHERE POINTS LEAK Starting an axis at a value other than 0 without marking it. That's fine if you label it clearly, but it's risky.

Part D (2 pts)

COACH: ASK YOURSELF

- Two far-apart points ON the line. Units of the slope: N ÷ (m/s²) = ?

A FULL-CREDIT ANSWER

Slope = $\Delta F/\Delta a \approx$ **1.2 kg** (N·s²/m = kg). Anything about 1.1–1.3 kg is consistent with the data.

Uses points from the best-fit line to compute the slope	1 pt
m ≈ 1.2 kg with correct units	1 pt

WHERE POINTS LEAK Leaving off units or giving N·s²/m without recognizing it as kg. Either is OK, but say kg.

THINK FURTHER If the track actually had a small amount of friction, how would the graph change? (It shifts up: there's a positive F-intercept equal to the friction force. The slope, and so the mass, doesn't change. That's why graphs beat single measurements.)

Question 4: Qualitative/Quantitative Translation

8 points

Satellite A orbits a planet of mass M in a circle of radius R. Satellite B, which has the same mass, orbits the same planet in a circle of radius 2R.

Part A (3 pts)

COACH: ASK YOURSELF

- What is the ONLY force on each satellite? What job does it do?
- Farther away: is that force stronger or weaker? And how much 'turning' does a bigger circle need at the same speed?

A FULL-CREDIT ANSWER

Gravity from the planet is the only force on each satellite, and it's what keeps each one moving in a circle (it provides the centripetal force). B is farther away, so the planet's gravitational pull on it is much weaker. A weaker inward force can only keep a satellite on a circle if it moves more slowly. B's circle is also larger, so it curves more gently, but the pull weakens faster than the curve flattens. So **B moves slower than A**.

Gravity is the (only) centripetal force	1 pt
Gravity is weaker at the larger distance	1 pt
Concludes B is slower, with reasoning that connects force to speed	1 pt

WHERE POINTS LEAK 'B is farther so it must go faster to cover more distance' mixes up speed and period. B does take longer, but because it's slower AND has farther to go.

Part B (3 pts)

COACH: ASK YOURSELF

- Set the gravitational force equal to m·(centripetal acceleration).
- Which mass cancels?

A FULL-CREDIT ANSWER

$\Sigma F = ma_c$ → $GMm/r^2 = mv^2/r$ → **$v = \sqrt{(GM/r)}$**. The satellite's mass cancels.

Gravitational force from universal gravitation	1 pt
Set equal to mv^2/r	1 pt
Correct expression	1 pt

WHERE POINTS LEAK Using mg with g = 9.8 for an orbit. Far from Earth, g isn't 9.8. Use GMm/r^2.

Part C (2 pts)

COACH: ASK YOURSELF

- $v \propto 1/\sqrt{r}$. What does r × 2 do to v?

A FULL-CREDIT ANSWER

v = √(GM/r) decreases as r increases, matching part A. v_B/v_A = √(R/2R) = **1/√2 ≈ 0.71**.

States that v decreases with r, consistent with part A	**1 pt**
Ratio 1/√2	**1 pt**

WHERE POINTS LEAK Ratio ½, which forgets the square root.

THINK FURTHER Geostationary satellites have a 24-hour period. Why must they all sit at the SAME altitude? (T depends only on r for a given planet, so one period gives exactly one radius.)

Error log: FRQ Set 2

For each miss or guess: what went wrong, using one of these types: **C** concept (didn't know the physics) · **R** representation (FBD/graph wrong) · **M** math/algebra · **F** FRQ wording (right idea, missing the words) · **T** time.

Q	Type	What I did vs. what I should have done	Re-try date / ✓

Score: _____ / 40 Sure-and-right: _____ Time used: _____